

# Xinbao Qiao

✆ +852-70141618 | ✉ xinbaoqiao@cuhk.edu.hk | 🏠 xinbaopedia.top | 🐙 GitHub | 🔗 LinkedIn | 📄 Google Scholar

## Education

**The Chinese University of Hong Kong** Ph.D., Information Engineering. Advisor: [Angela Yingjun Zhang](#). *Hong Kong. 2026~present*

**Research Areas** My research primarily studies lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted.

**Zhejiang University** M.Sc. in Artificial Intelligence. Advised by [Meng Zhang](#).

*Zhejiang. Sep 2022 ~ Dec 2025*

**Shandong University** B.Eng. in Communication Engineering.

*Shandong. Sep 2018 ~ Jul 2022*

## Experience

**Doctoral Research @ The Chinese University of Hong Kong**

*Hong Kong. 2026 ~ present*

- Study data-centric ML and AI-network co-design, including distributed Wasserstein barycenter methods for shared references without pooling raw data. **Paper #5**

**Research on Data-Centric ML Systems @ Zhejiang University**

*Zhejiang. Mar 2023 ~ Dec 2025*

- Developed soft-weighted unlearning methods that continuously correct data influence for fairness and robustness without binary erasure. **Paper #2**
- Designed Hessian-free online certified unlearning methods using recollected trajectory statistics instead of explicit Hessian inversion, supporting online deletion requests without second-order solves. **Paper #3**
- Studied exact and efficient random-forest unlearning in dynamic online environments, preserving predictive utility under incremental data updates. **Paper #4**

**Research on Trustworthy LLM Systems @ NUSRI CQ**

*Chongqing. Jun 2025 ~ Dec 2025*

- Analyzed low-resource sample selection in recursive synthetic-data training and proposed collaborative Wasserstein evaluation proxies. **Paper #1**
- Examined illusory pattern perception in LLMs and developed an interpretable framework for spurious inference behavior and misleading responses. **Paper #6**

## Open-Source Contributions and Services

- 🔗 **Research code releases**, public code for certified unlearning, soft-weighted unlearning, and model-collapse work.
- 🏠 **Xinbaopedia**, public academic homepage and wiki-style research archive with papers, figures, CV, and project notes.
- 📄 **Peer-Reviewing**, reviewer for ICML 2026, NeurIPS 2025 and 2026, ICLR 2025, AAAI 2025 and 2026, and IEEE TNNLS.

## Selected Publications

Asterisks (\*) denote co-first authorship. Accepted papers are listed before under-review manuscripts.

📄 📄 ICML 2026.

1. **When Sample Selection Bias Precipitates Model Collapse**

**Xinbao Qiao**, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang  
Low-resource verification can turn local sample selection into persistent tail pruning.

📄 📄 AAAI 2026.

2. **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness**

**Xinbao Qiao**, Ningning Ding, Yushi Cheng, Meng Zhang  
Soft-weighted corrective unlearning for non-binary fairness and robustness interventions.

📄 📄 ICLR 2025.

3. **Hessian-Free Online Certified Unlearning**

**Xinbao Qiao**, Meng Zhang, Ming Tang, Ermin Wei  
Certified unlearning without explicit Hessian inversion through recollected statistics.

📄 📄 ICLR 2025.

4. **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest**

Shurong Wang, Zhuoyang Shen, **Xinbao Qiao**, Tongning Zhang, Meng Zhang  
Exact and efficient random-forest unlearning in dynamic environments.

5. **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification**

**Xinbao Qiao**, Wenjing Yan, Ying-Jun Angela Zhang  
Barycentric multi-prototype classification for federated representation geometry.

*Under review.*

6. **Illusory Pattern Perception Drives Spurious Inference in Large Language Models**

Peihua Mai, Zhuoyan Shao, **Xinbao Qiao**, Meng Zhang, Xinyue Zhou, Yan Pang  
Analysis of spurious pattern perception and interpretable behavior in LLMs.

*Under review.*